

Midterm Exam 2 Solutions

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Answers 1 (T/F):

1. False.
2. False.
3. False.
4. False.
5. True.

Answers 2 (MCQ):

1. **Answer: (d)** $\alpha = [1 \ -1]^T / \sqrt{2}$, because MRC beamforming chooses $\alpha \propto \mathbf{h}^*$ to maximize $|\mathbf{h}^H \alpha|^2$ under a unit-norm constraint.
2. **Answer: (d)** Statistically the same, because both 2×1 TX MRC (with perfect CSIT) and 2×1 RX MRC realize an SNR proportional to $|h_1|^2 + |h_2|^2$ under the same power constraint.
3. **Answer: (b)** 2 independent complex symbols are sent over 2 time slots, so the Alamouti code rate is $R = 2/2 = 1$ symbol per channel use.
4. **Answer: (a)** In diversity mode the effective SNR with $N \times N$ channel scales with the number of antennas N^2 , while in multiplexing mode the number of parallel streams (and thus data rate) scales linearly with N .
5. **Answer: (d)** ToF (time-of-flight) resolution is determined by signal bandwidth, not by the number of antennas, so increasing N does not directly improve ToF resolution.

Solution 3: SIMO 1×2 Receiver Diversity with MRC

(a) Substitute y_1 and y_2 into the MRC combiner:

$$z = h_1^* y_1 + h_2^* y_2 = h_1^* (h_1 x + n_1) + h_2^* (h_2 x + n_2).$$

Expand:

$$z = \underbrace{(|h_1|^2 + |h_2|^2)}_{\tilde{h}} x + \underbrace{(h_1^* n_1 + h_2^* n_2)}_{\tilde{n}}.$$

So the effective input-output relation is

$$z = \tilde{h} x + \tilde{n}, \quad \tilde{h} = |h_1|^2 + |h_2|^2, \quad \tilde{n} = h_1^* n_1 + h_2^* n_2.$$

Since $n_1, n_2 \sim \mathcal{CN}(0, \sigma^2)$ are i.i.d. and independent of the channels,

$$h_1^* n_1 \sim \mathcal{CN}(0, |h_1|^2 \sigma^2), \quad h_2^* n_2 \sim \mathcal{CN}(0, |h_2|^2 \sigma^2),$$

and they are independent. Hence

$$\tilde{n} \sim \mathcal{CN}(0, (|h_1|^2 + |h_2|^2) \sigma^2).$$

The signal power at the output is

$$\mathbb{E}[|\tilde{h}x|^2] = |\tilde{h}|^2 \mathbb{E}[|x|^2] = (|h_1|^2 + |h_2|^2)^2 P.$$

and the noise power is

$$\mathbb{E}[|\tilde{n}|^2] = (|h_1|^2 + |h_2|^2) \sigma^2.$$

Therefore,

$$\text{SNR}_{\text{MRC}} = \frac{\mathbb{E}[|\tilde{h}x|^2]}{\mathbb{E}[|\tilde{n}|^2]} = \frac{(|h_1|^2 + |h_2|^2)^2 P}{(|h_1|^2 + |h_2|^2) \sigma^2} = \frac{(|h_1|^2 + |h_2|^2) P}{\sigma^2}.$$

(b) Given $|h_1|^2 = 0.2$, $|h_2|^2 = 1.8$, $P = 1$, $\sigma^2 = 0.1$:

(i) Using only antenna 1 (i.e., y_1 alone), the SNR is

$$\text{SNR}_1 = \frac{|h_1|^2 P}{\sigma^2} = \frac{0.2 \cdot 1}{0.1} = 2.$$

(In dB: $10 \log_{10}(2) \approx 3.0$ dB.)

(ii) Using only antenna 2 (i.e., y_2 alone), the SNR is

$$\text{SNR}_2 = \frac{|h_2|^2 P}{\sigma^2} = \frac{1.8 \cdot 1}{0.1} = 18.$$

(In dB: $10 \log_{10}(18) \approx 12.6$ dB.)

(iii) Using MRC, we have

$$\text{SNR}_{\text{MRC}} = \frac{(|h_1|^2 + |h_2|^2) P}{\sigma^2} = \frac{(0.2 + 1.8) \cdot 1}{0.1} = \frac{2.0}{0.1} = 20.$$

(In dB: $10 \log_{10}(20) \approx 13.0$ dB.)

So,

$$\text{SNR}_1 = 2, \quad \text{SNR}_2 = 18, \quad \text{SNR}_{\text{MRC}} = 20.$$

The MRC output SNR equals the *sum* of the individual SNRs:

$$\text{SNR}_{\text{MRC}} = \text{SNR}_1 + \text{SNR}_2.$$

Numerically, MRC is slightly better than the best single-antenna SNR (20 vs. 18). This shows diversity combining: even though antenna 1 alone is weak, it still contributes extra SNR when coherently combined, improving reliability beyond what the strongest antenna alone can provide.

- (c) In the correlated case $h_2 \approx h_1$ with $|h_1|^2 \approx |h_2|^2 \approx 1$, we still get some gain from MRC in terms of *average* SNR:

$$\text{SNR}_{\text{MRC}} \approx \frac{(|h_1|^2 + |h_2|^2)P}{\sigma^2} \approx \frac{(1+1)P}{\sigma^2} = 2 \frac{P}{\sigma^2} = 20,$$

which is about a 3 dB increase over a single antenna. However, because h_1 and h_2 fade in almost the same way (high correlation), deep fades tend to occur *simultaneously* on both branches. That means error events (when both channels are bad) are still likely, so the diversity benefit in terms of *outage probability* or *error slope* is limited.

In other words, MRC still gives an *array gain* (higher typical SNR), but not a large *diversity gain* in the presence of strong correlation. The key channel statistic that matters for diversity is how *independent* (or low-correlated) the fading on different antennas is: the more independent the branches, the larger the diversity order and the stronger the reduction in error probability at high SNR.

Solution 4: MIMO 2×2 with SVD-based Precoding

- (a) We have

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad \mathbf{x} = \mathbf{w}x_i,$$

where $\|\mathbf{w}\|^2 = 1$ and $\mathbb{E}[|x_i|^2] = P$. The receiver forms

$$\hat{x}_i = \mathbf{g}^H \mathbf{y}.$$

Substituting $\mathbf{y}$:

$$\hat{x}_i = \mathbf{g}^H (\mathbf{H}\mathbf{x} + \mathbf{n}) = \underbrace{(\mathbf{g}^H \mathbf{H} \mathbf{w})}_{\text{effective gain}} x_i + \underbrace{\mathbf{g}^H \mathbf{n}}_{\text{effective noise}}.$$

The effective scalar channel gain is $\mathbf{g}^H \mathbf{H} \mathbf{w}$ and the effective noise is $\mathbf{g}^H \mathbf{n}$.

- (b) The SVD of $\mathbf{H}$ is

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H, \quad \mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2),$$

with $\sigma_1 \geq \sigma_2 \geq 0$, and $\mathbf{U}, \mathbf{V}$ unitary. Let the columns of $\mathbf{U}$ be $\mathbf{u}_1, \mathbf{u}_2$ and those of $\mathbf{V}$ be $\mathbf{v}_1, \mathbf{v}_2$. We choose

$$\mathbf{w} = \mathbf{v}_1, \quad \mathbf{g} = \mathbf{u}_1.$$

Then the effective gain becomes

$$\begin{aligned} \mathbf{g}^H \mathbf{H} \mathbf{w} &= \mathbf{u}_1^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{v}_1 \\ &= (\mathbf{u}_1^H \mathbf{U}) \mathbf{\Sigma} (\mathbf{V}^H \mathbf{v}_1). \end{aligned}$$

Because $\mathbf{U}$ and $\mathbf{V}$ are unitary,

$$\mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2] \Rightarrow \mathbf{u}_1^H \mathbf{U} = [1 \ 0], \quad \mathbf{V} = [\mathbf{v}_1 \ \mathbf{v}_2] \Rightarrow \mathbf{V}^H \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Hence

$$\mathbf{g}^H \mathbf{H} \mathbf{w} = [1 \ 0] \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \sigma_1.$$

So the effective scalar channel is

$$\hat{x}_i = \sigma_1 x_i + \tilde{n}, \quad \tilde{n} \triangleq \mathbf{u}_1^H \mathbf{n}.$$

Since $\mathbf{U}$ is unitary, $\|\mathbf{u}_1\|^2 = 1$ and the noise covariance is preserved:

$$\tilde{n} = \mathbf{u}_1^H \mathbf{n} \sim \mathcal{CN}(0, N_0).$$

The signal term at the output is $\sigma_1 x_i$ with power

$$\mathbb{E}[|\sigma_1 x_i|^2] = \sigma_1^2 \mathbb{E}[|x_i|^2] = \sigma_1^2 P.$$

The noise power is N_0 . Thus the instantaneous SNR of the single stream is

$$\text{SNR}_{\text{MIMO},1\text{-stream}} = \frac{\sigma_1^2 P}{N_0}.$$

Note that σ_2 does not appear: only the strongest singular value matters for this one-stream SVD beamforming scheme.

- (c) The same symbol x_i is transmitted from both antennas along the strongest singular mode of the channel. We are not sending multiple independent streams, so there is *no* spatial multiplexing gain. Instead, multiple antennas are used coherently to focus power in the direction where the channel is strongest, yielding an *array gain* (higher effective channel gain σ_1 compared to a typical SISO gain). With coding over many channel uses, this also translates into an improved error performance similar to diversity-type gains, but the most direct interpretation here is transmit–receive array gain obtained by SVD-based beamforming and combining.

Solution 5: Coordinated downlink with two APs and two Clients

- (a) **Non-coordinated precoding.**

AP j transmits $x_j = m_j p_j$ for $j \in \{1, 2\}$. The received signals at clients A and B are

$$y_A = h_{A1}x_1 + h_{A2}x_2 + n_A = h_{A1}m_1p_1 + h_{A2}m_2p_2 + n_A,$$

$$y_B = h_{B1}x_1 + h_{B2}x_2 + n_B = h_{B1}m_1p_1 + h_{B2}m_2p_2 + n_B,$$

where n_A and n_B are the additive noise terms at clients A and B.

- (b) **Is interference-free transmission possible?**

We want each client to receive only its own symbol (up to a nonzero complex gain), i.e.,

$$y_A = g_A p_1 + n_A,$$

$$y_B = g_B p_2 + n_B,$$

for some nonzero complex numbers g_A and g_B .

Comparing with part (a), this requires that the cross-terms vanish *for all* symbol values p_1, p_2 :

$$h_{A2}m_2p_2 = 0 \quad \forall p_2,$$

$$h_{B1}m_1p_1 = 0 \quad \forall p_1.$$

Hence we must have

$$h_{A2}m_2 = 0, \quad h_{B1}m_1 = 0.$$

For generic channels with $h_{A2} \neq 0$ and $h_{B1} \neq 0$, the only way to satisfy these equations is

$$m_2 = 0 \quad \text{and} \quad m_1 = 0,$$

which means no data is transmitted. If we require m_1 and m_2 to be nonzero so that both p_1 and p_2 are sent, then interference-free transmission is *not* possible in this non-coordinated setting, except in the degenerate case where $h_{A2} = 0$ and $h_{B1} = 0$ (i.e., no cross-channels).

(c) **Cooperative linear precoding.**

Now each AP can transmit a linear combination of both symbols:

$$x_1 = m_{11}p_1 + m_{12}p_2, \quad x_2 = m_{21}p_1 + m_{22}p_2.$$

The received signals are

$$\begin{aligned} y_A &= h_{A1}x_1 + h_{A2}x_2 + n_A \\ &= h_{A1}(m_{11}p_1 + m_{12}p_2) + h_{A2}(m_{21}p_1 + m_{22}p_2) + n_A \\ &= (h_{A1}m_{11} + h_{A2}m_{21})p_1 + (h_{A1}m_{12} + h_{A2}m_{22})p_2 + n_A, \\ y_B &= h_{B1}x_1 + h_{B2}x_2 + n_B \\ &= h_{B1}(m_{11}p_1 + m_{12}p_2) + h_{B2}(m_{21}p_1 + m_{22}p_2) + n_B \\ &= (h_{B1}m_{11} + h_{B2}m_{21})p_1 + (h_{B1}m_{12} + h_{B2}m_{22})p_2 + n_B. \end{aligned}$$

In matrix form, define

$$\mathbf{y} = \begin{bmatrix} y_A \\ y_B \end{bmatrix}, \quad \mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}, \quad \mathbf{n} = \begin{bmatrix} n_A \\ n_B \end{bmatrix},$$

the channel matrix

$$\mathbf{H} = \begin{bmatrix} h_{A1} & h_{A2} \\ h_{B1} & h_{B2} \end{bmatrix},$$

and the precoding matrix

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}.$$

Then the received signal can be written compactly as

$$\mathbf{y} = \mathbf{H}\mathbf{M}\mathbf{p} + \mathbf{n}.$$

(d) **Zero-forcing precoder.**

We seek a precoding matrix $\mathbf{M}$ such that

$$\mathbf{y} = \mathbf{G}\mathbf{p} + \mathbf{n},$$

where $\mathbf{G}$ is diagonal and non-singular, i.e.,

$$\mathbf{H}\mathbf{M} = \mathbf{G} = \begin{bmatrix} g_A & 0 \\ 0 & g_B \end{bmatrix}, \quad g_A, g_B \neq 0.$$

If $\mathbf{H}$ is invertible, one convenient choice is

$$\mathbf{M} = \mathbf{H}^{-1} \mathbf{G}.$$

The simplest *zero-forcing* (ZF) choice is to take $\mathbf{G} = \mathbf{I}$, which gives

$$\boxed{\mathbf{M}_{\text{ZF}} = \mathbf{H}^{-1}}.$$

With this choice,

$$\mathbf{H} \mathbf{M}_{\text{ZF}} = \mathbf{H} \mathbf{H}^{-1} = \mathbf{I},$$

so each client receives only its own symbol (plus noise).

A ZF precoder of this form exists if and only if $\mathbf{H}$ is full rank (invertible), i.e.,

$$\det(\mathbf{H}) \neq 0.$$

(e) **Conceptual question.**

In the first scenario, each AP can only scale its *own* symbol, so there are not enough spatial degrees of freedom to cancel the interference at two different receivers for two simultaneously transmitted streams. In the second scenario, the APs can share both symbols and jointly choose a 2×2 precoding matrix $\mathbf{M}$, enabling them to implement the ZF condition $\mathbf{H} \mathbf{M}$ diagonal.

A key drawback of ZF precoding in practice is *noise amplification*: when the channel matrix $\mathbf{H}$ is ill conditioned (has very small singular values), $\mathbf{H}^{-1}$ contains large entries, which leads to large transmit power and makes the received SNR very sensitive to noise and channel estimation errors.

Solution 6: AoA and ToF Resolution in an OFDM-MIMO Sensing System

(a) **AoA resolution vs number of antennas.**

The AoA profile

$$P(\theta) = \left| \sum_{k=0}^{N-1} h_k e^{-jk\pi \cos \theta} \right|^2$$

behaves like the beam pattern of a length- N uniform linear array (ULA) with inter-element spacing $s = \lambda/2$. This is essentially the squared magnitude of a spatial DFT across antennas.

- *Main-lobe width*: As N increases, the aperture of the array increases (physical length $\approx (N-1)s$ grows). For a fixed wavelength and spacing, the main lobe of the array factor becomes *narrower*. Roughly, the main-lobe width in the spatial frequency domain (as a function of $\cos \theta$) scales on the order of $1/N$. Thus, doubling N approximately halves the beamwidth.
- *Minimum angular separation* $\Delta\theta_{\min}$: Two paths are resolvable in AoA if their angles produce two distinct peaks in $P(\theta)$, i.e., if their angular separation exceeds the main-lobe width. Therefore, the minimum resolvable angular separation satisfies

$$\Delta\theta_{\min} \propto \frac{1}{N}.$$

Increasing N improves AoA resolution: more antennas $\Rightarrow$ narrower beams $\Rightarrow$ smaller $\Delta\theta_{\min}$.

- **Bonus:** The phase term is $2\pi n \cos(\theta)/2$ (Assume $d = \lambda/2$)

FFT across a total N antennas will have resolution: $[\cos(\theta)/2]_{\min} = 1/N$, based on FFT principle.

Therefore, Resolution of $[\cos(\theta)]_{\min} = 2/N$

Hence proved.

(b) **ToF resolution vs bandwidth.**

- (i) In the delay (ToF) domain, the effective resolution is determined by the inverse of the signal bandwidth. A wideband signal of bandwidth B can be thought of as probing the channel with frequencies in a range of width B ; the corresponding time-domain impulse response is limited by this bandwidth. In particular, the smallest distinguishable delay difference is on the order of

$$\Delta\tau_{\min} \approx \frac{1}{B}.$$

Intuitively, higher bandwidth corresponds to finer ‘time granularity’, so we can separate paths that differ by smaller delays.

- (ii) If $\Delta\tau_{\min} \approx 1/B$, the corresponding *distance* (path-length) resolution is

$$\Delta d_{\min} = c \Delta\tau_{\min} \approx \frac{c}{B},$$

where c is the speed of light. This is the minimum separation in path length needed for two paths to appear as separate peaks in the delay (ToF) domain.

(c) **Numerical example.**

- (i) Only the AoA profile $P(\theta)$ For $N = 8$ antennas, the angular resolution is given by

$$\Delta \cos(\theta)_{\min} = \frac{2}{N} = 0.25 .$$

The difference is

$$|\cos(\theta_A) - \cos(\theta_B)| = |0.766 - (-0.5)| = 1.266 > \Delta \cos(\theta)_{\min} .$$

So, AP can separate paths A and B.

- (ii) Only the ToF (delay) profile Take $c = 3 \times 10^8$ m/s. For $B = 40$ MHz:

$$\Delta\tau_{\min} \approx \frac{1}{B} = \frac{1}{40 \times 10^6} = 5 \times 10^{-8} \text{ s} = 25 \text{ ns}.$$

Then

$$\Delta d_{\min} \approx c \Delta\tau_{\min} = (3 \times 10^8) \cdot (2.5 \times 10^{-8}) = 7.5 \text{ m}.$$

So with 40 MHz bandwidth, the system can resolve paths that differ in length by about 7.5 meters. Here the difference is $|5 - 7| = 2\text{m} < 7.5\text{m}$, so AP cannot separate paths A and B.

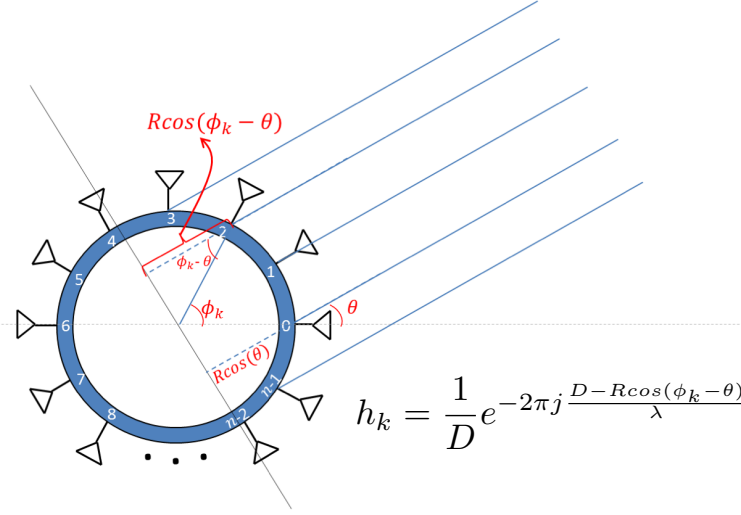


Figure 1: Circular antenna array of radius R with N antennas. Antenna k is at angle ϕ_k on the circle.

Solution 7: (ECSE 6560 only) Circular Antenna Arrays

We use the plane-wave (far-field) model from lecture: the field across the array is a complex exponential whose phase is proportional to the projection of the antenna position onto the propagation direction.

(a) Geometry and extra path length.

In the far field, the extra path length to antenna k relative to the array center is given by

$$\Delta \ell_k = -R \cos(\phi_k - \theta),$$

which comes from the geometry in Fig. 1.

(b) Channel to each antenna.

The total path length from the source to antenna k is

$$\ell_k \approx D + \Delta \ell_k = D - R \cos(\phi_k - \theta).$$

With free-space propagation, the complex baseband channel is proportional to $e^{-j2\pi \ell_k / \lambda}$. Ignoring amplitude variation across antennas and pulling out a common factor $1/D$, we obtain

$$\begin{aligned} h_k &= \frac{1}{D} e^{-j2\pi \ell_k / \lambda} \\ &= \frac{1}{D} e^{-j2\pi \frac{D - R \cos(\phi_k - \theta)}{\lambda}}, \end{aligned}$$

which is the desired expression.

(c) **Power pattern.**

To ‘beamform’ or ‘scan’ toward a candidate direction θ' , we apply a steering/compensation factor that removes the phase associated with a plane wave from θ' . From the same geometry, a wave from θ' produces an extra path $-R \cos(\phi_k - \theta')$; the corresponding phase is $-\frac{2\pi}{\lambda}(-R \cos(\phi_k - \theta'))$. Thus, a natural choice of steering factor is

$$f(R, \phi_k, \theta') = \exp\left(-j \frac{2\pi R}{\lambda} \cos(\phi_k - \theta')\right).$$

Then the beamformer output toward θ' is

$$\sum_{k=0}^{N-1} h_k f(R, \phi_k, \theta'),$$

and the power pattern is

$$P(\theta') = \left| \sum_{k=0}^{N-1} h_k f(R, \phi_k, \theta') \right|^2.$$

(d) **Why the maximum occurs at the true angle, and its value.**

Substitute the expression for h_k and $f(R, \phi_k, \theta')$:

$$\begin{aligned} h_k f(R, \phi_k, \theta') &= \frac{1}{D} e^{-j2\pi \frac{D-R \cos(\phi_k - \theta)}{\lambda}} e^{-j2\pi \frac{R \cos(\phi_k - \theta')}{\lambda}} \\ &= \frac{1}{D} e^{-j2\pi \frac{D}{\lambda}} e^{-j \frac{2\pi R}{\lambda} [\cos(\phi_k - \theta') - \cos(\phi_k - \theta)]}. \end{aligned}$$

Hence

$$\sum_{k=0}^{N-1} h_k f(R, \phi_k, \theta') = \frac{1}{D} e^{-j2\pi \frac{D}{\lambda}} \sum_{k=0}^{N-1} e^{-j \frac{2\pi R}{\lambda} [\cos(\phi_k - \theta') - \cos(\phi_k - \theta)]}.$$

When $\theta' = \theta$, the bracket is zero for all k , so every term in the sum has the *same* phase:

$$h_k f(R, \phi_k, \theta) = \frac{\lambda}{D} e^{-j2\pi \frac{D}{\lambda}} \quad \forall k.$$

Therefore

$$\sum_{k=0}^{N-1} h_k f(R, \phi_k, \theta) = \frac{1}{D} e^{-j2\pi \frac{D}{\lambda}} \sum_{k=0}^{N-1} 1 = \frac{N}{D} e^{-j2\pi \frac{D}{\lambda}}.$$

The corresponding power is

$$P(\theta' = \theta) = \left| \frac{N}{D} e^{-j2\pi \frac{D}{\lambda}} \right|^2 = \frac{N^2}{D^2}.$$

It is not asked in the question, but for your interest: For any $\theta' \neq \theta$, the phase term

$$-\frac{2\pi R}{\lambda} [\cos(\phi_k - \theta') - \cos(\phi_k - \theta)]$$

varies with k , so the N phasors no longer align. By the triangle inequality,

$$\left| \sum_{k=0}^{N-1} h_k f(R, \phi_k, \theta') \right| \leq \sum_{k=0}^{N-1} |h_k f(R, \phi_k, \theta')| = \frac{N}{D},$$

with equality only when all terms have the same phase, i.e., only when $\theta' = \theta$. Hence

$$\max_{\theta'} P(\theta') = P(\theta' = \theta) = \frac{N^2}{D^2}.$$

(e) **Comparison with linear arrays (front-back ambiguity).**

A symmetric uniform linear array (ULA) lying on a line has an inherent front-back ambiguity: the array response depends on $\cos \theta$, so directions $+\theta$ and $-\theta$ produce the same set of phase differences (up to a common phase), and the array cannot distinguish between them without additional information (e.g., polarization or pattern asymmetry).

For the circular array, the antennas surround the origin at different angles ϕ_k over $[0, 2\pi)$. A wave from azimuth θ and a wave from azimuth $\theta + \pi$ (the ‘back’ direction) give *very different* phase patterns around the ring:

$$\cos(\phi_k - \theta) \quad \text{vs.} \quad \cos(\phi_k - (\theta + \pi)) = -\cos(\phi_k - \theta).$$

These correspond to different steering vectors $\{h_k\}$ and hence different beamformer responses around the circle. As a result, the circular array does *not* suffer from the same front-back ambiguity as a symmetric ULA: it can uniquely resolve azimuth angles over the full 0-360° range.